



Raphaël TRAN-THANH

Ph.D. Student in Probability -
Stochastic processes

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🏠 Télécom Paris, Institut
Polytechnique de Paris

EDUCATION

► Ph.D. in Applied Mathematics

Télécom Paris - Palaiseau, France - Since Dec. 2025

Advisors: Laurent Decreusefond (Télécom Paris) and Nicolas Privault (NTU Singapore)

Key-words: Probability, point processes, determinantal processes, Gibbs processes, diffusion models.

► M.Sc. Applied Mathematics - Mathematics for Life Sciences

Institut Polytechnique de Paris - Palaiseau - From Sept. 2024 to Dec. 2025

Mathematics for Life Sciences track supervised by Sylvie Méléard.

Key-words: advanced stochastic processes, point processes, population dynamics, branching processes, epidemiology, genetic transmission.

► M.Sc. in Engineering - French "diplôme d'ingénieur"

Télécom Paris - Palaiseau - From Sept. 2022 to Dec. 2025

Random modeling (MACS) and *Signal processing for AI* (TSIA) study tracks.

Key-words: stochastic processes, Markov chains, point processes, Ito calculus, deep learning, time series, NLP.

RESEARCH EXPERIENCE

► Research Intern

Télécom Paris - Palaiseau, France - From Aug. 2025 to Nov. 2025

Advisor: Laurent Decreusefond (Télécom Paris)

Topic: Distance between solutions of stochastic differential equations and application to diffusion models.

► Research Intern

UQÀM - Montréal, Québec - From Mar. 2025 to Aug. 2025

Advisor: Hélène Guérin (UQÀM)

Topic: Propagation of chaos in a rank-dependent particle system.

OTHER RESEARCH PROJECTS

► Mating preferences and genetic transmission

École Polytechnique - Master's Thesis - From Sept. 2024 to July 2025

Advisors: Camille Coron (INRAE) and Diala Abu Awad (Université Paris-Saclay)

Topic: Markov models for genetic transmission with mating preferences.

TEACHING

► Martingales and Markov chains - APM_4MA02

Graduate level tutoring - Télécom Paris - Nov. 2025 to Feb. 2026

Content: Stochastic processes on general state spaces, asymptotic behavior, Markov kernels, strong Markov property.

SCIENTIFIC ACTIVITY

► **05/30/2026 – Welcome Day at IHES**

Institut des Hautes Études Scientifiques, Bures-sur-Yvette — Talk

Content: Bertrand's paradox and choosing 'at random'.

► **02/03/2026 – Trans Numériques in Rennes**

Couvent des Jacobins, Rennes — Short talk

Content: Point processes, generative models.

► **01/28/2026 – Rencontres MathTech**

Institut des Hautes Études Scientifiques, Bures-sur-Yvette — Talk

Content: Schrödinger bridge, diffusion models.

► **06/11/2026 – UQÀM Summer Seminar**

Université du Québec à Montréal, Montréal — Talk

Content: Population dynamics, gene transmission, Markov models.